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1. $f(x) = x^2 - 4x + 5 \quad x \leq 2 \quad g$

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2. $I, J \quad f : I \rightarrow J \quad g = f^{-1} : J \rightarrow I \quad f'(x) \neq 0 \quad x \in I$
$$g'(y) = \frac{1}{f'(g(y))} \quad (y \in J)$$

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3.

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(a) $\arcsin\left(-\frac{\sqrt{2}}{2}\right)$ (b) $\arccos\left(-\frac{\sqrt{3}}{2}\right)$ (c) $\arctan \sqrt{3}$

$$1. y = f(x) = (x - 2)^2 + 1 \quad x \leq 2 \quad x - 2 \leq 0$$

$$x - 2 = -\sqrt{y - 1},$$

$$g(y) = 2 - \sqrt{y - 1}$$

$$f \quad (-\infty, 2] \quad [1, \infty) \quad g \quad [1, \infty) \quad (-\infty, 2]$$

$$2. y \in J \quad f(g(y)) = y \quad y$$

$$f'(g(y)) g'(y) = 1$$

$$f'(g(y)) \neq 0 \quad f'(g(y))$$

$$g'(y) = \frac{1}{f'(g(y))}$$

$$3. (a) \quad -\frac{\pi}{4} \quad (b) \quad \frac{5\pi}{6} \quad (c) \quad \frac{\pi}{3}$$

$$\sin(-\pi/4) = -\sqrt{2}/2 \quad \cos(5\pi/6) = -\sqrt{3}/2 \quad \tan(\pi/3) = \sqrt{3} \quad [-\pi/2, \pi/2] [0, \pi] (-\pi/2, \pi/2)$$